HELPYARD | LOCAL-FIRST PROJECT

Project Upgrade & Launch Plan

A staged, testable path from the current single-page dashboard and Windows bridge to a reliable local developer workspace.

Plan number: 1
Prepared: 17 October 2026
Project baseline: HTML/CSS/JavaScript dashboard + PowerShell localhost bridge

Current release gate: NOT READY FOR PUBLIC LAUNCH

> **Verified blockers found and fixed:** the bridge contained a PowerShell parser error and duplicate authorization helpers; the Ollama route also streamed responses while the dashboard expected a single JSON reply. These defects are corrected. Full cross-browser validation and the remaining release gates below still need to pass before claiming launch readiness.

This plan separates the repair executed now from subsequent hardening and feature upgrades. No step treats an unrun test as a pass.

1. Objectives and boundaries

- Keep the current lightweight HTML/CSS/JavaScript frontend and localhost Windows bridge as the initial product architecture.
- Make startup, browser connection, errors, and test outcomes observable and repeatable.
- Keep privileged operations local, token-protected, explicitly approved when risky, and unavailable to remote hosts.
- Prioritize reliability and security before adding 3D visuals, project indexing, editor features, or public distribution.

> **Public exposure warning:** the bridge is a local workstation-control service, not a public web API. Never expose its port to the internet or a LAN. A public demo must use a separate mock/sandbox backend with no access to personal files, processes, or system controls.

2. Execution order and completion criteria

Step	Priority	Implementation	Done when
1. Repair baseline	P0 / executed	Fix the malformed authorization condition; remove duplicate authorization code; route all protected requests through the single valid helper.	PowerShell AST parser reports zero errors; expected-token request passes and invalid/missing token is rejected. Syntax and token checks passed in Windows PowerShell 5.1.
2. Verify bridge API	P0 / partial	Run bridge on loopback; check ping, auth failures, health/stats, chat serialization, and malformed or unknown requests. Confirm graceful stop and useful errors.	Repeatable local smoke tests pass on Windows PowerShell 5.1 and PowerShell 7 if supported. Ping/auth and Ollama chat passed; the full route matrix remains open.
3. Stabilize dashboard	P0	Guard corrupt/unavailable browser storage; show connection and provider failures; remove silent error swallowing; validate required DOM state and saved settings.	No uncaught page errors; offline mode remains usable; each failure is visible and recoverable.
4. Harden bridge policy	P0	Validate request size and JSON, origin/host, route and arguments; apply least privilege and approval on the bridge; constrain process/service operations; add bounded audit records.	Negative tests prove rejected requests do not change system state; bridge remains loopback-only.
5. Reproducible setup	P1	Add a user-mode launcher, clear prerequisites, startup diagnostics, port/token guidance, health check, and uninstall/recovery steps.	Fresh Windows account can start and stop the app from documented steps without admin rights unless a specific approved feature requires them.
6. Cross-browser QA	P1	Test current stable Edge, Chrome, and Firefox on Windows at desktop and narrow widths; exercise navigation, settings, reconnect, local AI absent/present, and accessibility basics.	Recorded matrix contains browser/version, OS, scenario, result, and evidence; no unsupported-browser claims.
7. Release security review	P1	Review token lifecycle, CORS, local-network exposure, secrets, dangerous actions, logging, and threat model; run dependency and static checks.	No unresolved critical/high findings; public-facing build uses an isolated mock backend.
8. Project intelligence	P2	Implement explicit project-folder authorization, safe file enumeration, ignores/size limits, manifest detection, and local search before AI context.	Tests confirm secrets/generated files are excluded and paths cannot escape the approved workspace.
9. AI and edit workflow	P2	Add Ollama health/streaming, source citations, proposed diffs, explicit review, backups, apply/reject/revert, and approved test execution.	AI cannot write or execute without user review; failures and exit codes are shown accurately.

Step	Priority	Implementation	Done when
10. Public pilot	P3	Publish a demo build with simulated data only, privacy notice, support channel, rollback, versioned release notes, and opt-in feedback.	External test does not grant access to any developer machine or live bridge.

3. Detailed work packages

Phase A — Baseline repair and smoke tests (P0)

1. Keep the authorization logic singular and parseable.
2. Parse the complete bridge file before each test run.
3. Start it with a test token on a free loopback port.
4. Verify authorized ping returns the expected JSON; verify absent/wrong token is rejected.
5. Check CORS preflight and invalid origins; confirm the listener is bound only to 127.0.0.1.
6. Record exact PowerShell version and outcomes.

Phase B — Frontend resilience and accessibility (P0/P1)

1. Provide visible loading, connected, offline, and retry states for each service-backed panel.
2. Handle invalid localStorage data without preventing initial render; avoid silently losing settings.
3. Expose errors through the UI and diagnostic log without leaking credentials.
4. Support keyboard focus, semantic labels, adequate contrast, reduced motion, and responsive layouts.
5. Verify no external network access is required for the local-only baseline.

Phase C — Security and operations (P0/P1)

1. Document the localhost-only security boundary and prevent LAN/public binding.
2. Require confirmation server-side for destructive or privileged operations; validate IDs and service names.
3. Limit command execution capability; do not treat an authenticated browser token as a safe command policy.
4. Bound input/output and request duration; return stable JSON errors and meaningful status codes.
5. Test hostile origins, invalid JSON, oversized inputs, unknown routes, bad tokens, and rejected approvals.

Phase D — Release and public pilot (P1/P3)

1. Provide an easy local launcher and clean shutdown instructions.
2. Run and publish a browser/OS/version matrix; distinguish verified, unsupported, and untested cases.
3. Run security review and resolve release-blocking findings.
4. For public evaluation, package a demo mode with synthetic metrics and no workstation bridge or user data.
5. Use opt-in telemetry only, with documented collection and retention; default to none.

Phase E — Product upgrades (P2)

1. Add approved-folder project explorer and safe project metadata discovery.
2. Integrate local model status and source-grounded assistant context.
3. Add reviewable code diffs and rollback-safe patch application.
4. Add approved tasks with timeouts, cancellation, output, and exit status.
5. Only then consider a lightweight 3D command center or broader GitHub/MCP features.

4. Test and release matrix

Area	Required checks	Release evidence
Bridge syntax	PowerShell 5.1 parser; PowerShell 7 parser when supported	Zero parser errors, recorded runtime versions
Bridge behavior	Valid/invalid auth, loopback bind, ping, API errors, OPTIONS/origin, shutdown	Automated smoke-test output and exit codes
Frontend	Fresh state, corrupt state, offline bridge, navigation, chat failure, reconnect	Browser console clean; visible user-facing errors
Browsers	Stable Edge, Chrome, Firefox on Windows; desktop and mobile-width viewports	Versioned results; list any browser not actually tested
Safety	Unapproved destructive action, non-admin restricted action, invalid input	All denied without system state change
Public demo	Network inspection, synthetic data, no machine-control bridge	Independent demo deployment and rollback verification

5. Current status and launch decision

Executed in this work: fixed the PowerShell parse error in `Assert-Authorized`, removed the duplicate helper, and updated the request handler to call the unified authorization helper. Fixed Ollama response serialization by forcing a non-streaming response; the dashboard returned `READY` in a live chat test. Windows PowerShell 5.1 parsing is clean. In the integrated Chrome 150 browser on Windows 11, all 12 navigation views loaded; project, process, service, system-stat, terminal, and local-model flows were exercised without page errors.

Still required before launch: full negative/edge-case route coverage, offline and recovery UX review, direct UI testing in additional target browsers (Firefox is not installed in this environment), permission/security hardening, and a separate safe public demo mode.

Decision: do not expose the current bridge publicly. Proceed with local verification and P0 fixes; approve a public pilot only after all P0/P1 gates pass using a mock backend.